



Batterjee Medical College
For Science and Technology

Intracellular Signaling pathways

Dr. Hader I. Sakr

Associate Professor, Medical Physiology

بِسْمِ اللَّهِ الرَّحْمَنِ
الرَّحِيمِ



وَاتَّقُوا اللَّهَ^{صلى} *
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Learning outcomes:

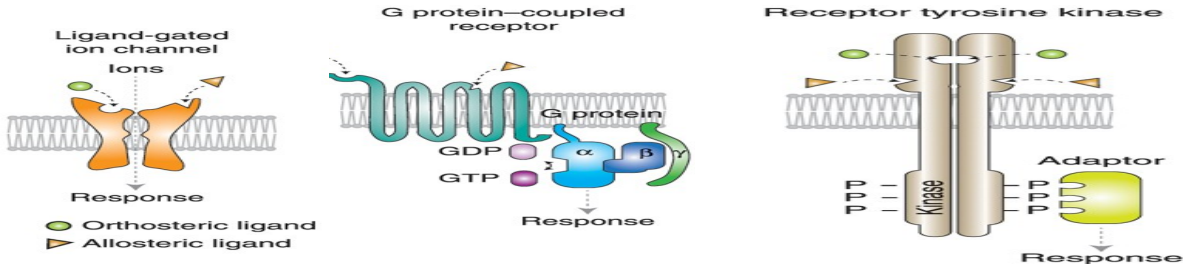
By the end of the lecture, the student will be able to:

- **Explain hormonal receptors intracellular signaling pathways**
- **Describe Channels activity alternation**
- **Identify G-protein coupled receptors**
- **Discuss enzyme activation and transcription modulation**



Hormones receptors intracellular signaling pathways

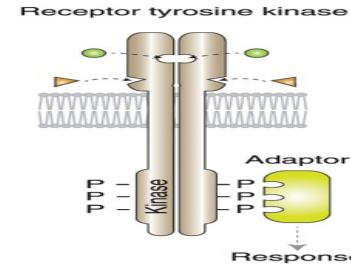
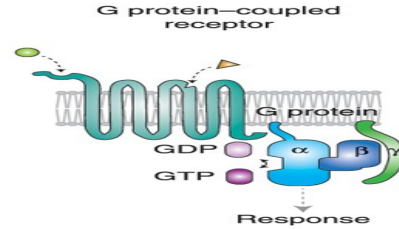
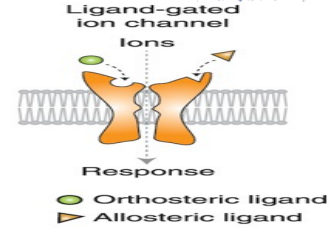
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Hormones receptors intracellular signaling pathways

☐ The mechanisms by which chemical messengers exert their IC effects may be through:

- A. Opening of certain ion channels (trans-membrane).
- B. G-Protein activation (trans-membrane).
- C. Altering enzyme activity within the cell (trans-membrane or cytoplasmic).
- D. Activation of transcription.



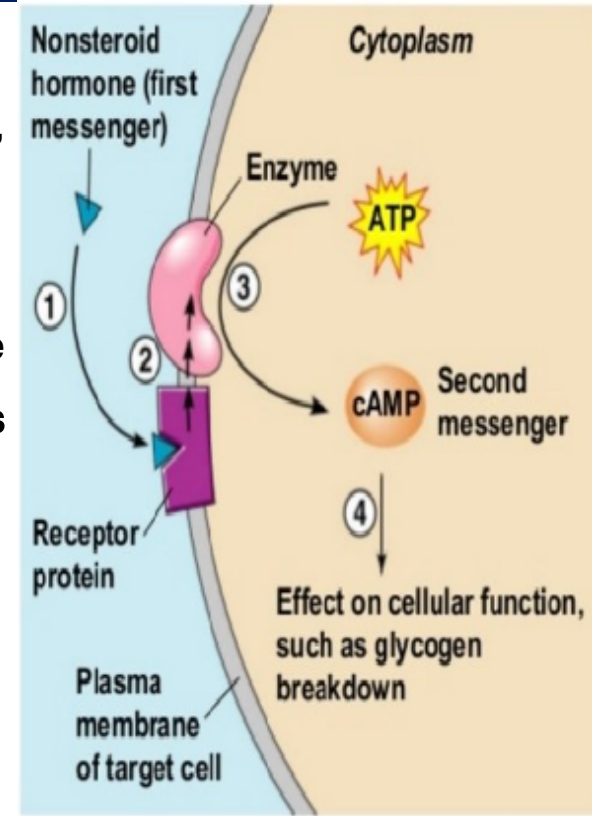
The resulting cell-signaling pathways provide:

- 1. **Amplification** (same) and **distribution** (different) of the effect.
- 2. **Regulation** and **feedback** control.

Hormones receptors intracellular signaling pathways

1. Trans-membrane receptors

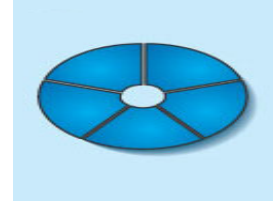
- ☐ For **lipid-insoluble** hormones (protein & peptide, catecholamines, serotonin, melatonin).
- ☐ In or on the surface of the cell membrane.
- ☐ Combination of the hormone with the cell membrane receptors will regulate the activity of **target proteins** (enzymes or ion channels) either:
 - Directly
 - Through coupling with G proteins.



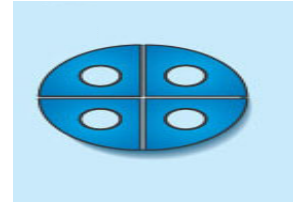
Hormones receptors intracellular signaling pathways

A. Opening of certain ion channels.

a In pentameric 'ligand-gated' cation (K^+) or anion (Cl^-) channels (acetylcholine or GABA), the single pore is formed by 5 identical or structurally similar subunits.



b, Aquaporin water channels are tetramers, again with one pore per subunit (ADH).



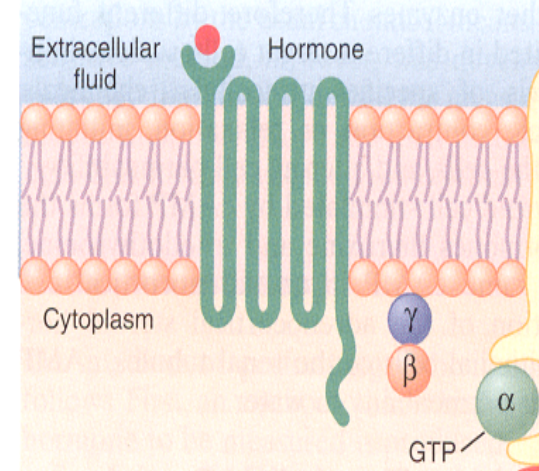
B. G- Protein activation (G-protein coupled receptors (GPCR):

- [?] Termed so because of their ability to bind guanosine nucleotide.
- [?] When the activating signal reaches the GPCR, they exchange GDP for GTP.
- [?] The GTP-protein complex brought about the effect.
- [?] Inherent **GTPase** activity of the protein converts GTP to GDP, restoring the inactive **resting** state to **terminate** its activity.

Hormones receptors intracellular signaling pathways

[?] Heterotrimeric G proteins

- Couple surface receptor (GPCR) to ion channel or intracellular catalytic unit forming the 2nd messenger.
- Over 1000 receptors are coupled; each has 7 trans-membrane loops.
- Activation of a single GPCR can result in 1, 10 or more active G proteins (**amplification**).
- Made up of 3 subunits: α , β & γ .
- Subunit α subunits are divided into three subfamilies: s, l and q, and a single GPCR can couple to either one or more families



$G\alpha_s$	$G\alpha_i$	$G\alpha_q$
$\uparrow$ Adenyl cyclase $\downarrow$		Phospholipase C_β

Hormones receptors intracellular signaling pathways

[?] Each G protein activates several downstream effectors:

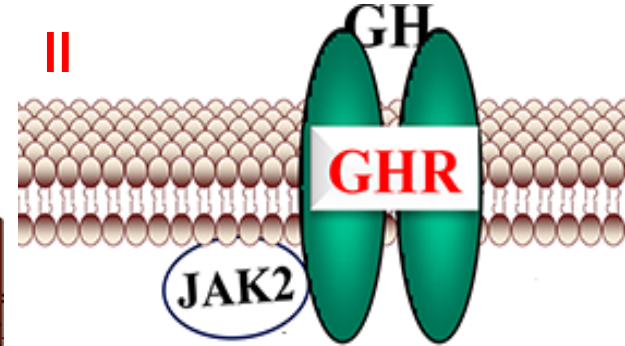
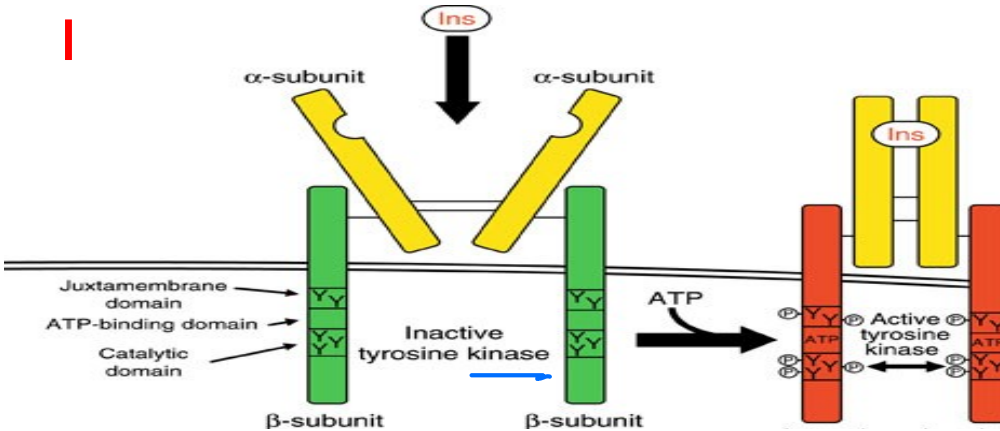
- $G\alpha_s$ **stimulates** adenylyl cyclase and **increases** levels of cAMP.
- $G\alpha_i$ **inhibits** adenylyl cyclase and **lowers** cAMP levels.
- $G\alpha_q$ family bind to and **activate** phospholipase C (PLC).



[?] The $G\beta\gamma$ subunits dimer activates many signaling molecules, including phospholipases, ion channels and lipid kinases.

Hormones receptors intracellular signaling pathways

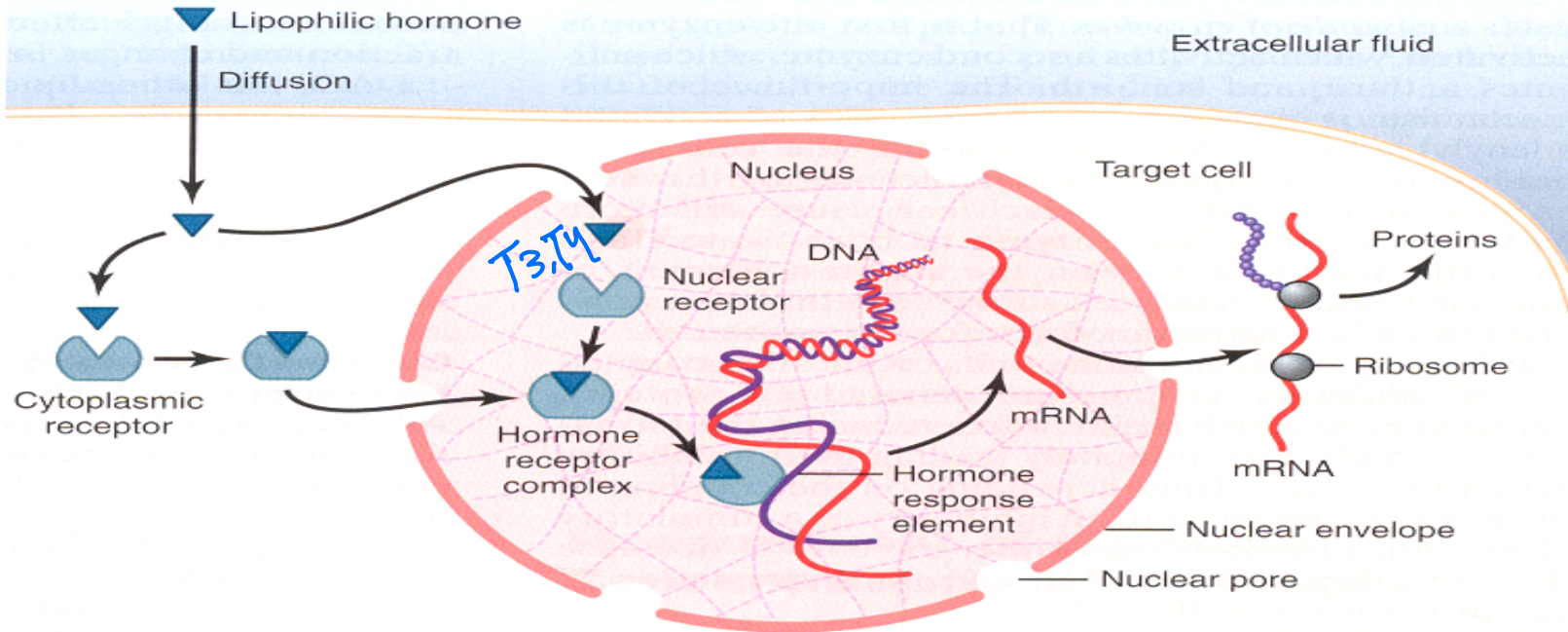
C. Enzyme-Linked Hormone Receptors:



- When activated, they function **directly** as enzymes (I, part of the **transmembrane** receptor) or are closely **associated** with enzymes (II, the H-R complex activates **cytoplasmic** enzyme).
- They have their hormone-binding site on the **outside** of the cell membrane and their **catalytic** (inherited, I) or enzyme-binding (cytoplasmic, II) site on the **inside**.

Hormones receptors intracellular signaling pathways

2. Cytoplasmic receptors (Steroids, Lipid-soluble)



Hormones receptors intracellular signaling pathways

- ☐ They can cross the cell membrane (**lipid-soluble**) and interact with **cytoplasmic** (or **nuclear**) receptors rather than in the cell membrane.
- ☐ The activated **H-R complex** then binds with the **hormone response element** (HRE) within a specific gene regulatory sequence (**promoter**), altering its transcription & mRNA formation.
- ☐ There are 3 distinct pathways for 1ry messenger to alter transcription:
 - ~~1.~~ Activation of **cytoplasmic** latent transcription factor.
 - ~~2.~~ Activation of **cytoplasmic** protein kinase cascade → migrate to the **nucleus** → phosphorylate a **nuclear** latent transcription factor.
 - ~~3.~~ Binding to cytoplasmic or nuclear receptor with **different** effects.

Thank You!



Conclusion

- Hormone receptors may be trans-membrane, cytoplasmic or nuclear.
- Trans-membrane receptors act through ion channels, GPCR or enzyme-linked.
- Ion channels have different configurations.
- GPCRs are classified according to the G-protein into small and heterotrimeric.
- Hormone-activated enzymes may be part of the receptor or cytoplasmic.

Case scenario

A 24-year-old man is bitten by a snake whose venom contains a toxin that irreversibly locks open the nicotinic acetylcholine receptor at the neuromuscular junction. This receptor belongs to which structural class of trans-membrane receptor?

- A. Pentameric ligand-gated ion channel
- B. Heterotrimeric G-protein-coupled receptor
- C. Enzyme-linked receptor with intrinsic catalytic activity
- E. Cytoplasmic (nuclear) receptor

References

Guyton and Hall, 13th edition. Unit II(6).

Ganong's review of medical physiology 25th ed. Section I(5).

Thank You